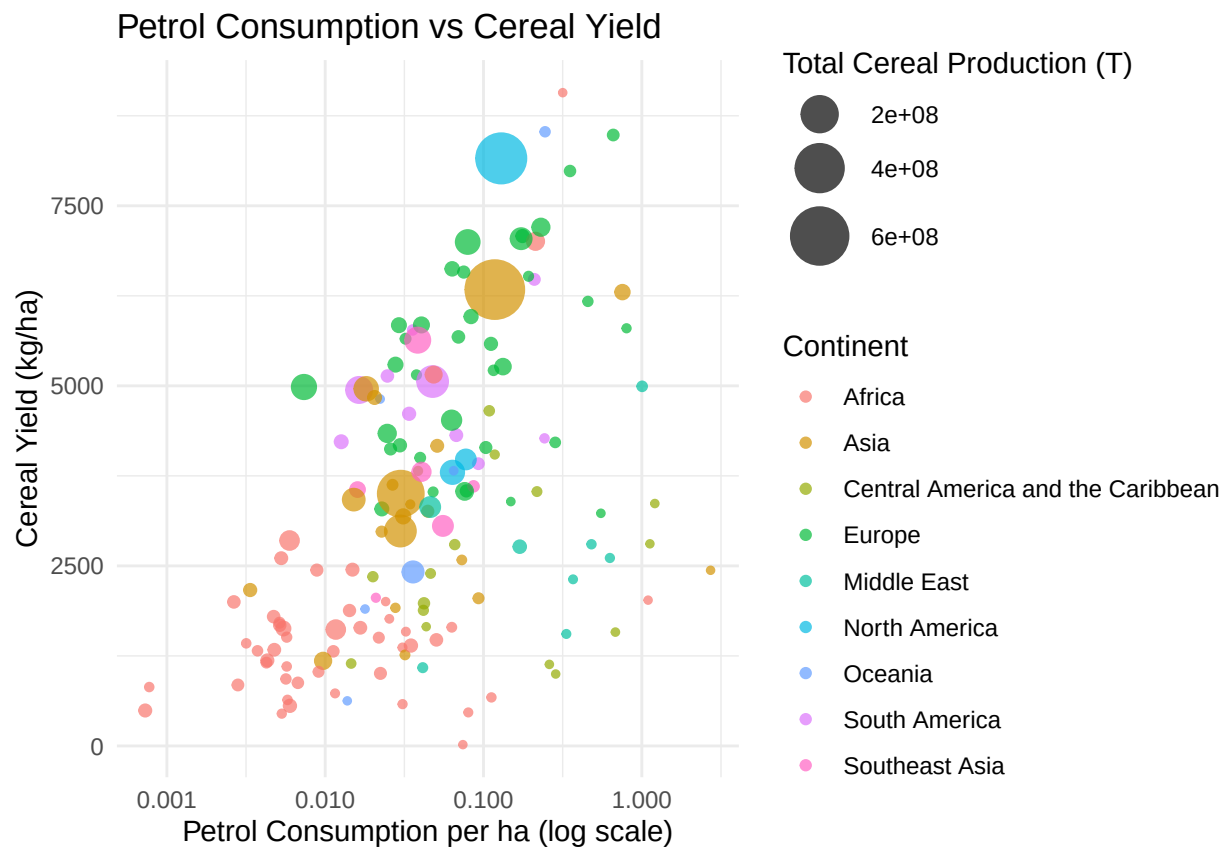


Question5

2025-06-18

Can we identify global patterns in agricultural performance and energy-water usage among countries ?

overview



```
library(tidyverse)
library(reshape2) # pour melt

##
## Attaching package: 'reshape2'

## The following object is masked from 'package:tidyr':
##
## smiths
```

```

# Étape 1 : Min-max normalization des deux variables
df_violin <- merged_df %>%
  select(Continent, cereal_yield_kg_ha, agricultural.Water.withdrawal.per.HA) %>%
  mutate(
    cereal_yield_scaled = (cereal_yield_kg_ha - min(cereal_yield_kg_ha, na.rm = TRUE)) /
      (max(cereal_yield_kg_ha, na.rm = TRUE) - min(cereal_yield_kg_ha, na.rm = TRUE))

    water_use_scaled = (agricultural.Water.withdrawal.per.HA - min(agricultural.Water.withdrawal.per.HA, na.rm = TRUE)) /
      (max(agricultural.Water.withdrawal.per.HA, na.rm = TRUE) - min(agricultural.Water.withdrawal.per.HA, na.rm = TRUE))
  ) %>%
  select(Continent, cereal_yield_scaled, water_use_scaled)

# Étape 2 : passer en format long pour ggplot
df_long <- df_violin %>%
  pivot_longer(cols = c(cereal_yield_scaled, water_use_scaled),
    names_to = "Variable",
    values_to = "Value")

# Étape 3 : Plot en facette violon par continent et variable
ggplot(df_long, aes(x = Continent, y = Value, fill = Variable)) +
  geom_violin(alpha = 0.7, position = position_dodge(0.9), trim = FALSE) +
  scale_fill_manual(
    values = c("cereal_yield_scaled" = "#1f77b4", "water_use_scaled" = "#ff7f0e"),
    labels = c("Cereal Yield", "Water Use")
  ) +
  labs(
    title = "Distribution of Normalized Cereal Yields and Water Use by Continent",
    y = "Normalized Value (Min-Max)",
    x = "Continent",
    fill = "Variable"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

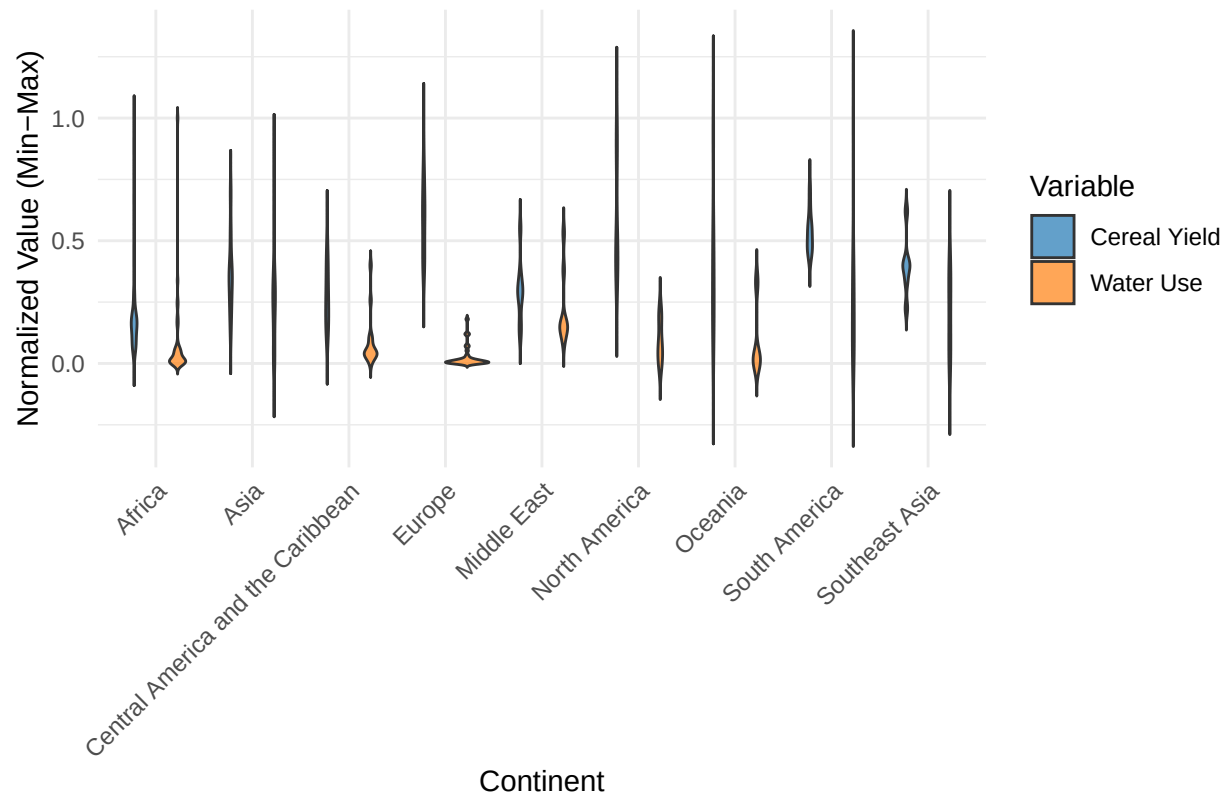
```

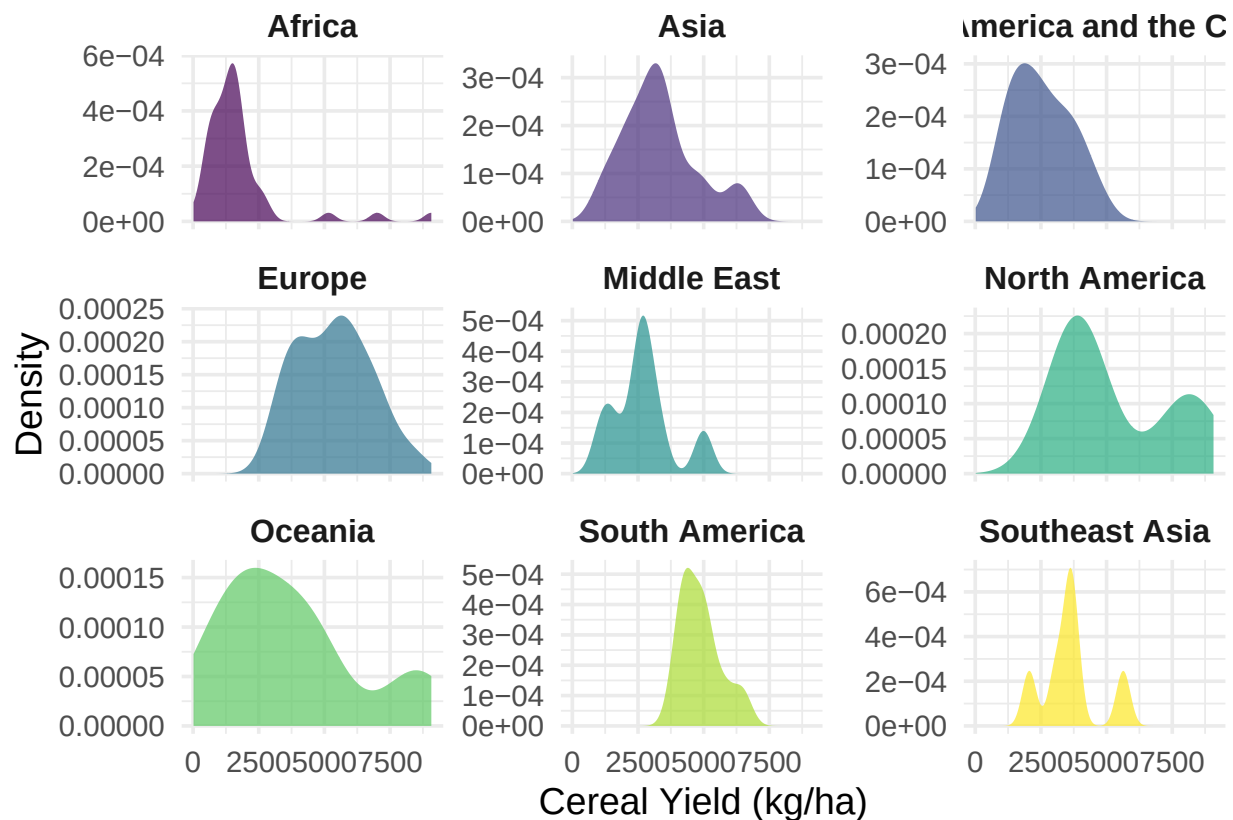
```

## Warning: Removed 3 rows containing non-finite outside the scale range
## ('stat_ydensity()').

```

Distribution of Normalized Cereal Yields and Water Use by Continent



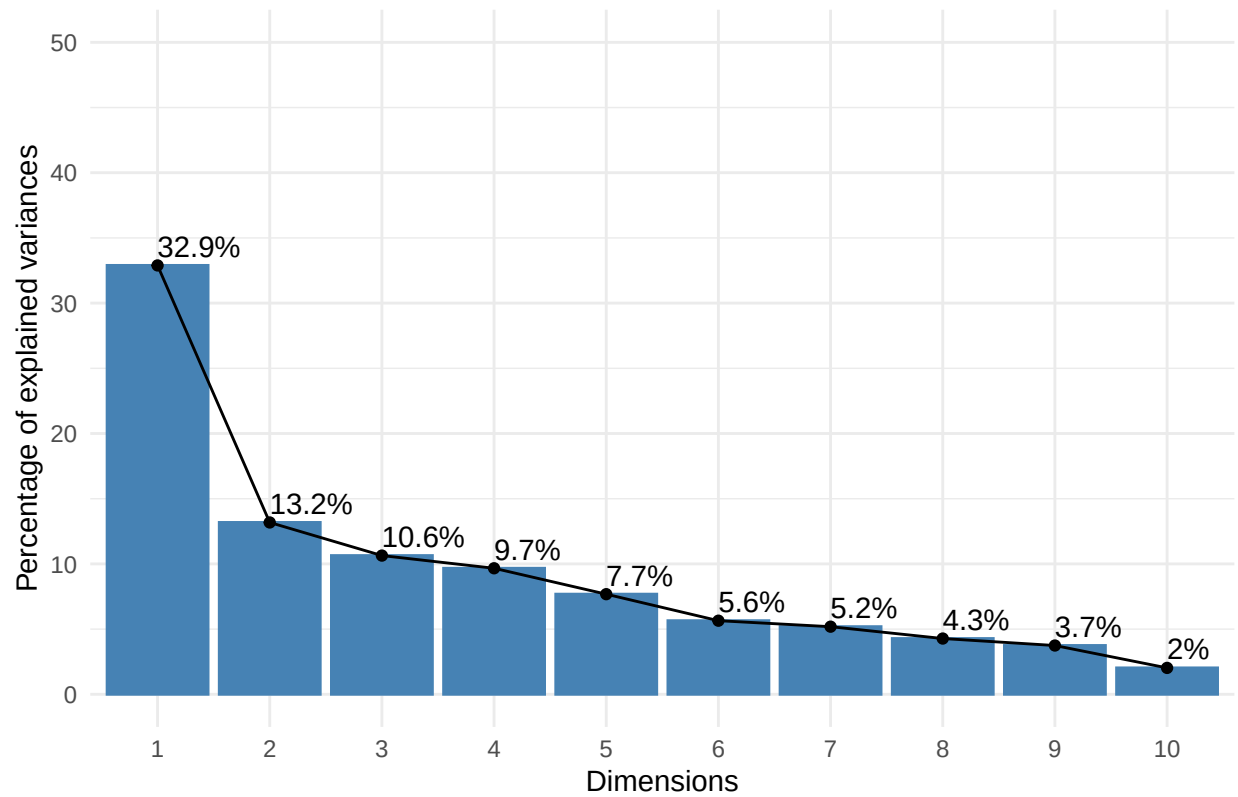


Principal Component Analysis (PCA)

In this analysis, we performed a Principal Component Analysis (PCA) on a set of standardized (per hectare of arable land) agricultural, environmental, and input-related variables across countries. The variables include several yield measures (e.g., cereal, maize, fruit, pulses, sugar crops), energy consumption per hectare (coal, petrol, natural gas), agricultural water withdrawal per hectare, fertilizer input (NPK, herbicides, insecticides), agricultural production value per hectare, and the share of irrigated cropland.

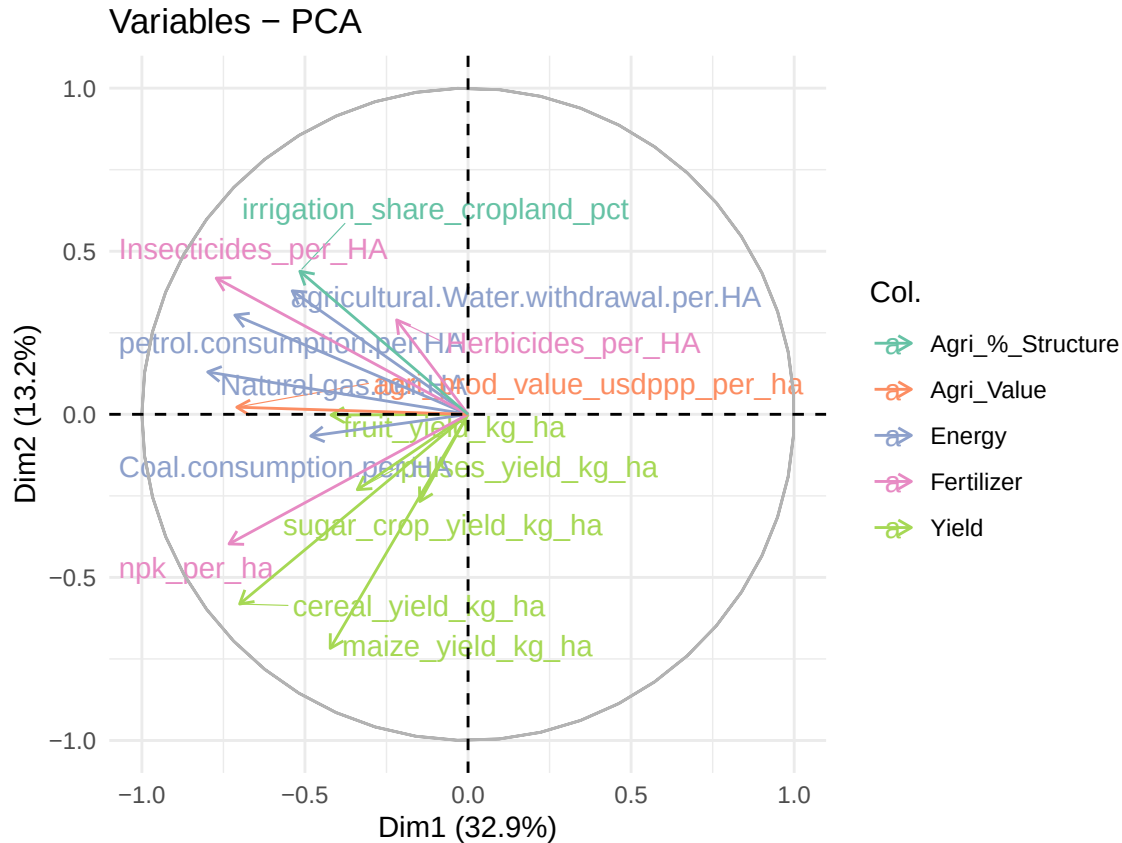
These variables were grouped into thematic categories—Yield, Energy, Fertilizer, Water, Agricultural Structure, Value, and Production—to better understand the multidimensional structure of agricultural efficiency and resource use across countries.

Scree Plot: Variance Explained by Principal Components



The scree plot below shows the percentage of total variance explained by each principal component. The first two principal components explain approximately 33% and 13% of the variance, respectively, accounting for nearly 46% of the total variability.

While the elbow in the scree plot is not very pronounced (with a slight inflection point after the first component), we proceed by focusing on the first two dimensions to explore the main patterns in the data.



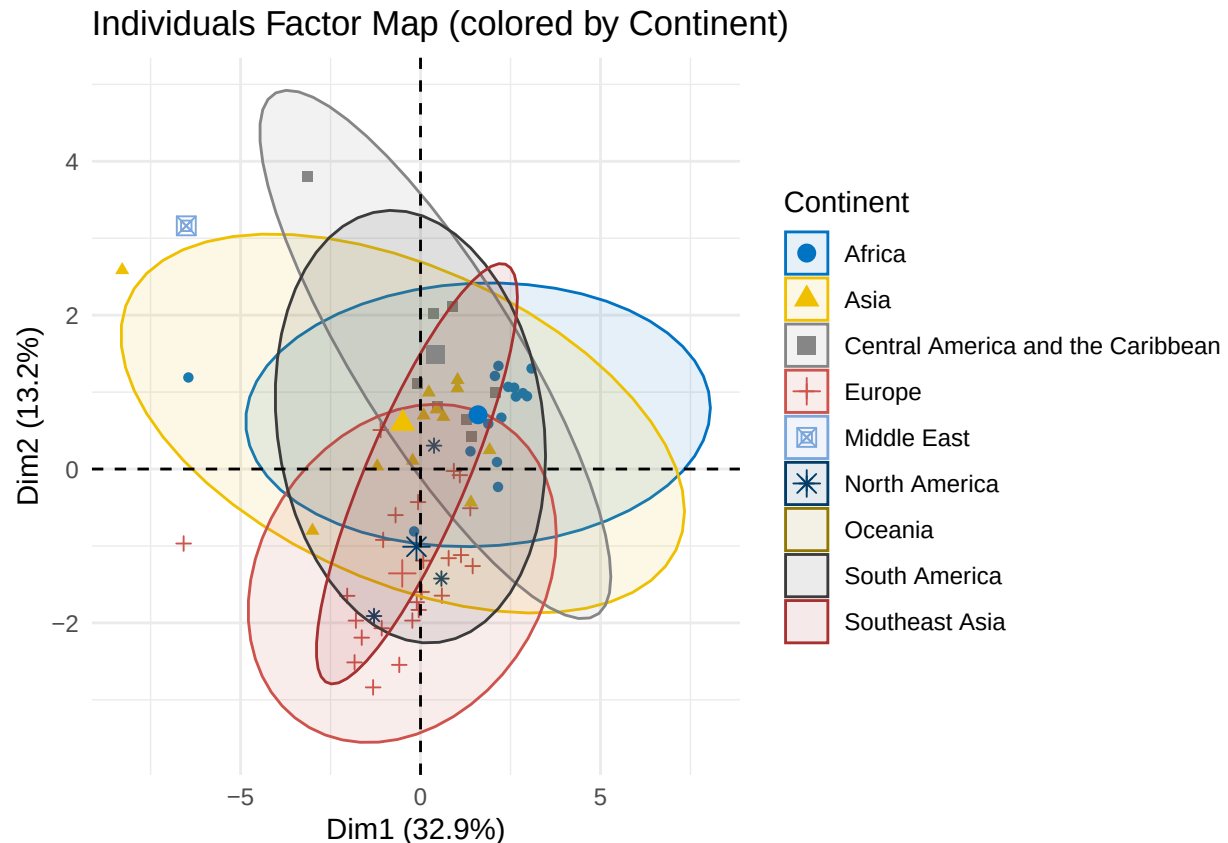
This plot shows how each variable contributes to the two main principal components. The direction and length of each arrow represent the contribution and correlation of the variable with the components.

Variables related to agricultural yields cluster in the bottom-left quadrant, suggesting they contribute strongly to Dimension 2. In contrast, variables associated with energy consumption (coal, petrol, gas) are mostly orthogonal to the yield variables and point toward the upper-left quadrant, indicating a strong contribution to Dimension 1. This suggests that yield and energy consumption are not strongly correlated in this dataset.

Similarly, most input-related variables (insecticides, herbicides) align with energy variables, with the exception of NPK fertilizers, which point in the same direction as the yield variables. This may indicate that NPK use is more closely associated with crop productivity than other inputs.

Interestingly, water consumption variables align with energy usage rather than with yields, suggesting that higher water withdrawal does not necessarily translate into higher agricultural performance at a global scale.

The variable representing the average agricultural production value per hectare (USD PPP) points in an intermediate direction. A reasonable interpretation is that energy consumption is more reflective of general development level (e.g., high energy use in oil-exporting countries with low crop yields), rather than of agricultural performance. In contrast, countries with developed agricultural systems and access to irrigation might achieve both higher yield and higher value-added production per hectare. Hence, the average production value captures both productivity and the diversity of economically valuable crops.



This graph displays countries projected on the first two principal components, colored by continent. The dimensions reflect contrasting agricultural and resource consumption profiles:

Dimension 1 (32.9% of variance): Primarily separates countries by energy and water consumption. Countries on the left exhibit higher usage of petrol, coal, gas, and agricultural water, while countries on the right have lower consumption or other distinguishing features.

Dimension 2 (13.2% of variance): Opposes countries with high agricultural yields and NPK fertilizer use (bottom) to those with higher energy/water inputs relative to output (top).

A continental reading highlights several regional patterns:

Bottom-Left quadrant (high yields, high NPK, low energy use): Concentration of European and Southeast Asian countries, suggesting more efficient or yield-oriented agricultural systems with moderate input reliance.

Upper-Left quadrant (high energy/water use, low yield): Includes several North American and Oceanian countries, hinting at higher resource intensity with less emphasis on yield efficiency in the observed variables.

Right-hand side (lower input usage): Many African and Asian countries are projected here, likely reflecting lower overall resource consumption or differing agricultural structures.

Central area: Countries from the Middle East, Central America, and the Caribbean tend to cluster near the origin, indicating more balanced or heterogeneous profiles not dominated by any one component.

Lastly, the agricultural production value per hectare, pointing in an intermediate direction, may align more with the bottom-left quadrant—supporting the idea that high yields and targeted input use (like NPK) contribute to higher economic returns per hectare, especially in developed or export-oriented systems.